INTERNATIONAL GCSE MATHEMATICS A · LINEAR 4MA1 HIGHER

Top 20 Questions for 4MA1

The questions that are coming

Twenty exam-style questions on the ideas that appear in almost every 4MA1 Higher paper — statistics, probability, algebra, trigonometry, circles, functions, vectors and calculus. One question per page, full solutions at the back.



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20	83	100%	1
QUESTIONS	MARKS	ORIGINAL QUESTIONS	QUESTION PER PAGE

THE TWENTY

1 Histogram — Draw and Estimate
2 Histogram — Read It Back
3 Tree Diagram — Without Replacement
4 Probability with Algebra
5 Algebraic Fraction — Simplify
6 Equation with Algebraic Fractions
7 Bounds — Suitable Accuracy
8 Surds
9 Cosine Rule — Angle and Area
10 3D Trigonometry — Pyramid
11 Cumulative Frequency
12 Reverse Percentage
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16 Functions
17 Maximum Volume
18 Vectors — Straight-Line Proof
19 Change the Subject
20 Trigonometric Graph

BOTTOM LINE Do each question in full, on its page, without looking. Then mark it with the solutions at the back (page 22). Every question you needed the solution for: do it again tonight.

1 Histogram — Draw and Estimate

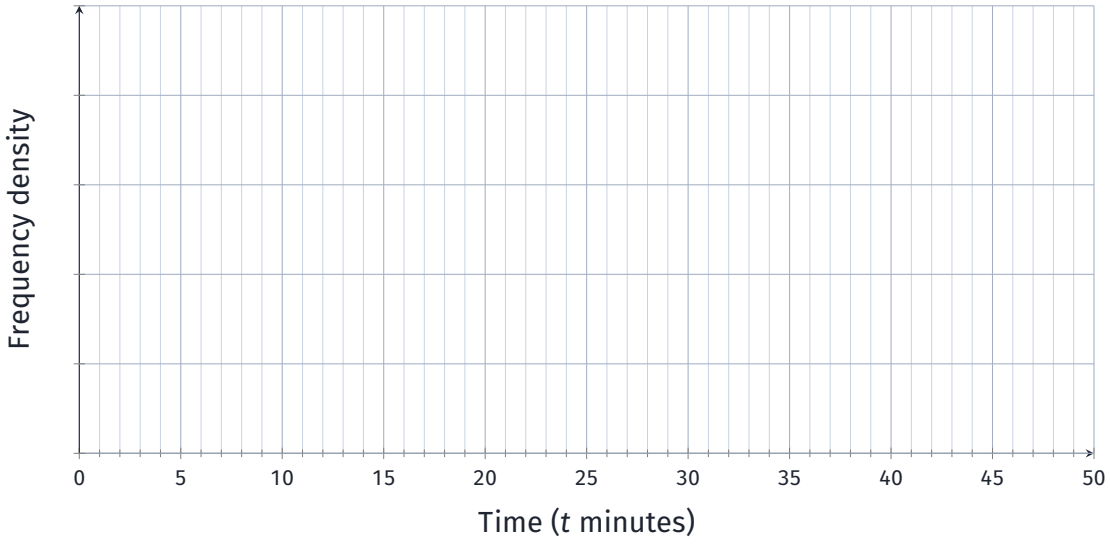
5 MARKS

Why it matters: a histogram is in almost every 4MA1 Higher paper.

The table gives information about the times, t minutes, that 110 customers waited.

Time (t minutes)	Frequency
$0 < t \leq 5$	10
$5 < t \leq 15$	32
$15 < t \leq 20$	24
$20 < t \leq 30$	28
$30 < t \leq 50$	16

(a) On the grid, draw a histogram for this information. **(3)**



(b) Work out an estimate for the number of customers who waited between 12 and 25 minutes. **(2)**

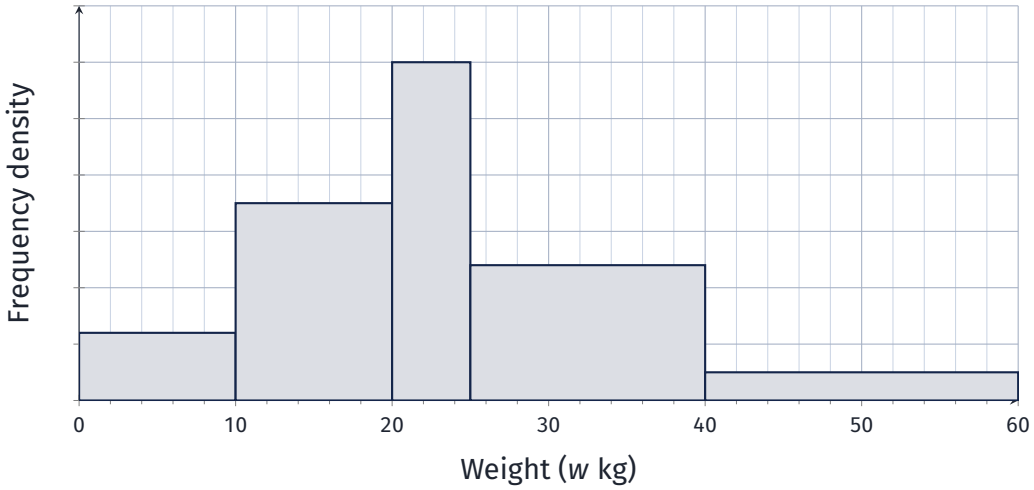
WORKING

2 Histogram — Read It Back

4 MARKS

Why it matters: "no scale on the axis" is the twist examiners love.

The histogram shows information about the weights, w kg, of some parcels.
12 parcels have a weight in the class $0 < w \leq 10$.



(a) Work out the total number of parcels. **(2)**

(b) Work out an estimate for the fraction of the parcels that weigh more than 30 kg. **(2)**

WORKING

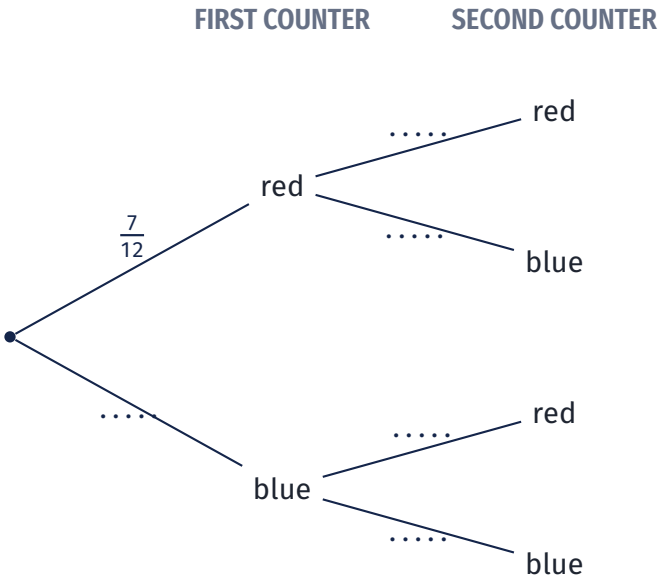
3 Tree Diagram — Without Replacement

4 MARKS

Why it matters: a tree diagram is in almost every paper.

There are 7 red counters and 5 blue counters in a bag.
Lena takes at random a counter from the bag and does **not** put it back. She then takes at random a second counter.

(a) Complete the probability tree diagram. **(2)**



(b) Work out the probability that the two counters are different colours. **(2)**

WORKING

4 Probability with Algebra

5 MARKS

Why it matters: a probability that turns into a quadratic — a favourite last-page question.

A bag contains x red counters and 5 blue counters only. There are more red counters than blue counters.
Omar takes at random two counters from the bag, without replacement.
The probability that he takes one counter of each colour is $\frac{10}{21}$

(a) Show that $x^2 - 12x + 20 = 0$ **(3)**

(b) Find the number of red counters in the bag. **(2)**

WORKING

5 Algebraic Fraction — Simplify

3 MARKS

Why it matters: factorise everything first, then cancel.

Simplify fully $\dfrac{2x^2 + 7x - 15}{4x^2 - 9}$ **(3)**

WORKING

6 Equation with Algebraic Fractions

4 MARKS

Why it matters: fractions + quadratic formula in one question.

Solve $\frac{3}{x+1} + \frac{2}{x-2} = 1$

Give your solutions correct to 3 significant figures. Show clear algebraic working. **(4)**

WORKING

7 Bounds — Suitable Accuracy

5 MARKS

Why it matters: a 5-mark bounds question appears most years.

A metal block has mass 450 g, correct to the nearest 10 g.
The volume of the block is 62.4 cm^3, correct to 1 decimal place.

(a) Work out the upper bound for the density of the block. Give your answer to 3 s.f. **(3)**

(b) By considering bounds, write down the density to a suitable degree of accuracy. Give a reason for your answer. **(2)**

WORKING

8 Surds

4 MARKS

Why it matters: a surd question appears in most papers.

(a) Expand and simplify $(2\sqrt{3} - 1)^2$ **(2)**

(b) Show that $\dfrac{12}{\sqrt{7} - 2}$ can be written in the form $a + b\sqrt{7}$ where a and b are integers. **(2)**

WORKING

9 Cosine Rule — Angle and Area

5 MARKS

Why it matters: non-right-angled triangles are in every paper.

The diagram shows triangle ABC.
$AB = 7$ cm, $AC = 9$ cm, $BC = 12$ cm.

(a) Work out the size of angle BAC. Give your answer correct to 1 decimal place. **(3)**

(b) Work out the area of the triangle. Give your answer correct to 3 significant figures. **(2)**

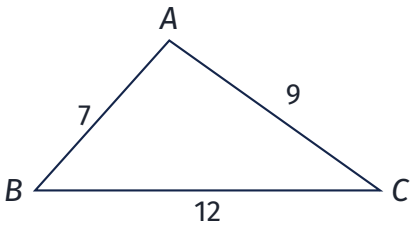


Diagram NOT accurately drawn

WORKING

10 3D Trigonometry — Pyramid

3 MARKS

Why it matters: 3D trigonometry is in most papers and often left blank.

VABCD is a pyramid with a square base *ABCD* of side 10 cm.
The vertex *V* is vertically above the centre *M* of the base, and *VM* = 12 cm.

Work out the size of the angle between the edge *VA* and the base *ABCD*.
Give your answer correct to 1 decimal place. **(3)**

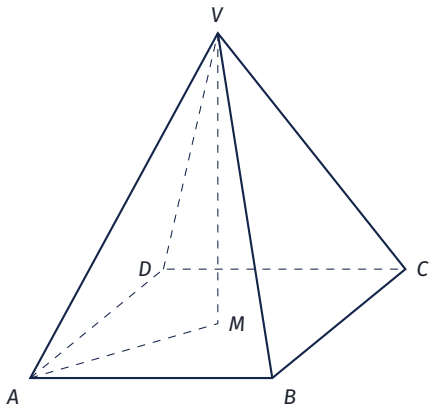


Diagram NOT accurately drawn

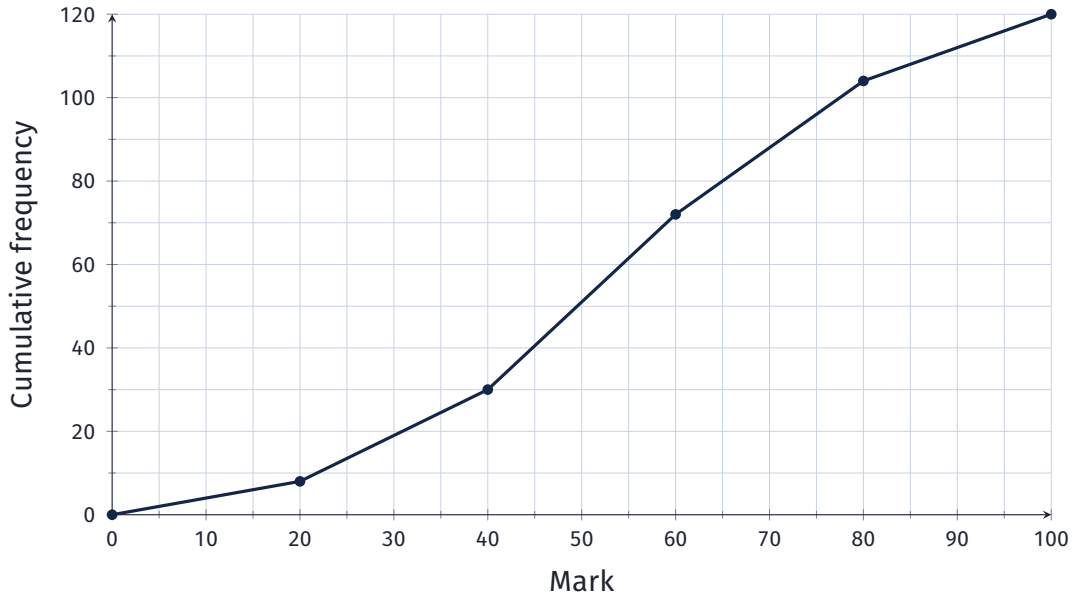
WORKING

11 Cumulative Frequency

5 MARKS

Why it matters: cumulative frequency appears in most papers.

The cumulative frequency graph shows the marks of 120 students in a test.



(a) Use the graph to find an estimate for the median mark. **(1)**

(b) Use the graph to find an estimate for the interquartile range. **(2)**

(c) The pass mark is 70. Work out an estimate for the number of students who passed. **(2)**

WORKING

12 Reverse Percentage

3 MARKS

Why it matters: a “before the sale” question is in most papers.

In a sale, all prices are reduced by 15%.
The sale price of a coat is $91.80
Work out the price of the coat before the sale. **(3)**

WORKING

13 Circle Theorems

4 MARKS

Why it matters: every angle needs a reason.

A, *B* and *C* are points on a circle, centre *O*.
TA is the tangent to the circle at *A*.
Angle *TAB* = 58°

(a) Find the size of angle *ACB*. Give a reason. **(2)**

(b) Find the size of angle *OBA*. Give reasons. **(2)**

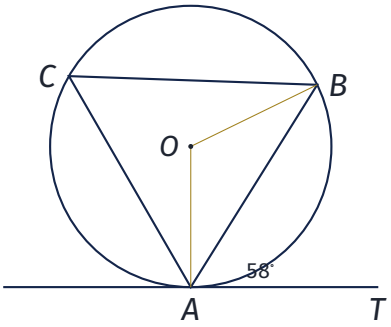


Diagram NOT accurately drawn

WORKING

14 Similar Solids

3 MARKS

Why it matters: length factor first, then square or cube it.

Two bottles are mathematically similar. The smaller bottle has height 12 cm and the larger bottle has height 18 cm.

(a) The volume of the smaller bottle is 400 cm^3. Work out the volume of the larger bottle. **(2)**

(b) The surface area of the larger bottle is 540 cm^2. Work out the surface area of the smaller bottle. **(1)**

WORKING

15 Simultaneous — Line and Curve

5 MARKS

Why it matters: a 5-mark line-and-curve question is common on Paper 2.

Solve the simultaneous equations $y = x + 1$ $x^2 + 2y^2 = 22$

Show clear algebraic working. **(5)**

WORKING

16 Functions

5 MARKS

Why it matters: composite and inverse functions are in most papers.

$f(x) = 2x + 3$ $g(x) = x^2$ $h(x) = \dfrac{x + 1}{x - 2}$

(a) Find $fg(-2)$ **(1)**

(b) Find $h^{-1}(x)$ **(2)**

(c) Solve $gf(x) = 25$ **(2)**

WORKING

17 Maximum Volume

5 MARKS

Why it matters: calculus on a real shape — a typical late question.

A square sheet of card has side 24 cm. A square of side x cm is cut from each corner and the sides are folded up to make an open box.

(a) Show that the volume of the box is $V = x(24 - 2x)^2$ **(1)**

(b) Find the value of x that gives the maximum volume, and the maximum volume. **(4)**

WORKING

18 Vectors — Straight-Line Proof

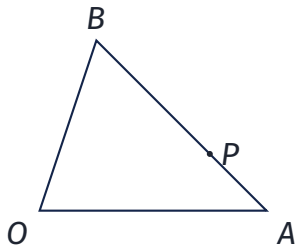
4 MARKS

Why it matters: the vectors proof is usually worth 4–6 marks.

$\vec{OA}$ = **a** and $\vec{OB}$ = **b**.
P is the point on AB such that $AP : PB = 1 : 2$
$\vec{OQ} = 2\mathbf{a} + \mathbf{b}$

(a) Find $\vec{OP}$ in terms of **a** and **b**. **(2)**

(b) Show that O, P and Q lie on a straight line. **(2)**



WORKING

19 Change the Subject

4 MARKS

Why it matters: the letter appears twice — collect, then factorise.

Make k the subject of $p = \dfrac{3k + 2}{k - 5}$ **(4)**

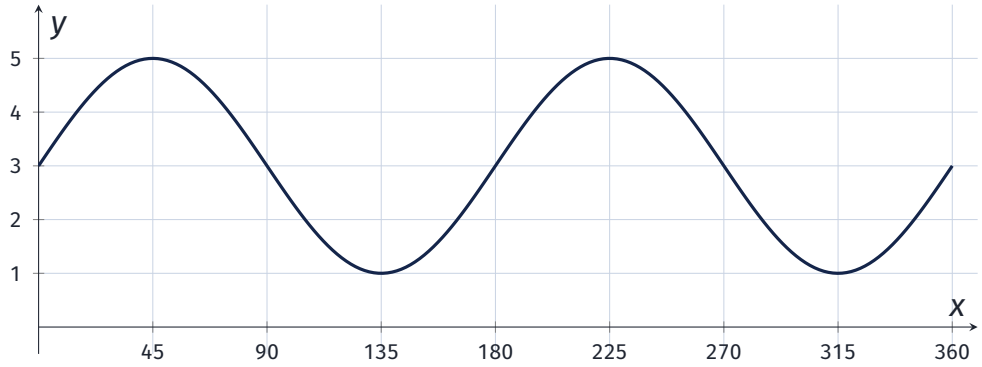
WORKING

20 Trigonometric Graph

3 MARKS

Why it matters: read a, b and c from the graph.

The graph of $y = a\sin(bx^\circ) + c$ for $0 \le x \le 360$ is shown.



Find the value of a, of b and of c. **(3)**

WORKING

20 Full Solutions

Mark your work line by line. A method mark is for the right step even if the arithmetic slips — so check your steps, not only the answer.

1 Histogram — draw and estimate

1 **FREQUENCY DENSITIES**
$10 \div 5 = 2$, $32 \div 10 = 3.2$, $24 \div 5 = 4.8$, $28 \div 10 = 2.8$, $16 \div 20 = 0.8$. Bars of these heights, widths = the classes.

2 **12 TO 25 MINUTES**
$(15 - 12) \times 3.2 = 9.6$, 15–20: 24, $(25 - 20) \times 2.8 = 14$

ANSWER (b) $9.6 + 24 + 14 = 47.6 \approx 48$ customers

2 Histogram — read it back

3 **FIND THE SCALE**
The 0–10 bar is 12 parcels and has width 10, so its height is FD = 1.2: one big square = 1 unit.

4 **EACH BAR: FD × WIDTH**
$1.2 \times 10 = 12$, $3.5 \times 10 = 35$, $6 \times 5 = 30$, $2.4 \times 15 = 36$, $0.5 \times 20 = 10$

5 **MORE THAN 30 KG**
$(40 - 30) \times 2.4 = 24$ plus 10 = 34

ANSWER (a) 123 (b) $\frac{34}{123}$

3 Tree diagram — without replacement

6 **BRANCHES**
First: red $\frac{7}{12}$, blue $\frac{5}{12}$. After red: red $\frac{6}{11}$, blue $\frac{5}{11}$. After blue: red $\frac{7}{11}$, blue $\frac{4}{11}$.

7 **DIFFERENT COLOURS = RB + BR**
$\frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11} = \frac{35}{132} + \frac{35}{132} = \frac{70}{132}$

ANSWER (b) $\frac{35}{66}$

4 Probability with algebra

8 **ONE OF EACH COLOUR = RB + BR**

$$\frac{x}{x+5} \times \frac{5}{x+4} \times 2 = \frac{10x}{(x+5)(x+4)}$$

9 **SET EQUAL AND SIMPLIFY**

$$\frac{10x}{(x+5)(x+4)} = \frac{10}{21} \Rightarrow 21x = x^2 + 9x + 20 \Rightarrow x^2 - 12x + 20 = 0$$

10 **SOLVE**

$(x - 2)(x - 10) = 0$, $x = 2$ or $x = 10$. More red than blue (5), so $x = 10$.

ANSWER (b) 10 red counters

5 Algebraic fraction — simplify

11 **FACTORISE**

$2x^2 + 7x - 15 = (2x - 3)(x + 5)$; $4x^2 - 9 = (2x - 3)(2x + 3)$

ANSWER $\dfrac{x+5}{2x+3}$

6 Equation with algebraic fractions

12 **MULTIPLY BY** $(x + 1)(x - 2)$

$3(x - 2) + 2(x + 1) = (x + 1)(x - 2)$

13 **EXPAND AND COLLECT**

$5x - 4 = x^2 - x - 2 \Rightarrow x^2 - 6x + 2 = 0$

14 **FORMULA**

$$x = \frac{6 \pm \sqrt{36 - 8}}{2} = 3 \pm \sqrt{7}$$

ANSWER $x = 5.65$ or $x = 0.354$

7 Bounds — suitable accuracy

15 **BOUNDS**

$445 \le m < 455$; $62.35 \le V < 62.45$

16 **UB DENSITY = UB MASS ÷ LB VOLUME**

$455 \div 62.35 = 7.2975\ldots$

17 **LB DENSITY = LB MASS ÷ UB VOLUME**

$445 \div 62.45 = 7.1257\ldots$

ANSWER (a) 7.30 g/cm^3 (b) 7 g/cm^3 — both bounds round to 7 to 1 s.f.

8 Surds

18 **(A)** $(2\sqrt{3}-1)(2\sqrt{3}-1)$

$12 - 2\sqrt{3} - 2\sqrt{3} + 1 = 13 - 4\sqrt{3}$

19 **(B) CONJUGATE**

$$\frac{12(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)} = \frac{12(\sqrt{7}+2)}{7-4} = 4(\sqrt{7}+2)$$

ANSWER (a) $13 - 4\sqrt{3}$ (b) $8 + 4\sqrt{7}$

9 Cosine rule — angle and area

20 **ANGLE A IS OPPOSITE THE 12**

$$\cos A = \frac{7^2 + 9^2 - 12^2}{2(7)(9)} = \frac{-14}{126}$$

21 **INVERSE COS**

$A = 96.379\ldots^\circ$ (obtuse — cos is negative)

22 **AREA**

$\frac{1}{2}(7)(9)\sin 96.379^\circ = 31.30\ldots$

ANSWER (a) 96.4° (b) $31.3\ \text{cm}^2$

10 3D trigonometry — pyramid

23 **HALF THE DIAGONAL**

$AC = \sqrt{10^2 + 10^2} = 14.142\ldots$, so $AM = 7.071\ldots$

24 **TRIANGLE AMV, RIGHT ANGLE AT M**

$$\tan\theta = \frac{12}{7.071\ldots}$$

ANSWER 59.5°

11 Cumulative frequency

25 **MEDIAN AT 60**

read across from 60: about 54

26 **QUARTILES AT 30 AND 90**

LQ = 40, UQ ≈ 71, IQR ≈ 31

27 **PASS MARK 70**

read up from 70: about 88 students scored 70 or less

ANSWER (a) 54 (b) 31 (c) 120 − 88 = 32

12 Reverse percentage

28 **SALE PRICE IS 85%**

$91.80 \div 0.85$

ANSWER \$108

13 Circle theorems

29 **(A)**

$ACB = 58^\circ$ — alternate segment theorem

30 **(B)**

$AOB = 2 \times 58 = 116^\circ$ (centre = 2× circumference); triangle OAB is isosceles ($OA = OB$, radii)

ANSWER (a) 58° (b) $(180 - 116) \div 2 = 32^\circ$

14 Similar solids

31 **LENGTH FACTOR**

$18 \div 12 = 1.5$

ANSWER (a) $400 \times 1.5^3 = 1350\ \text{cm}^3$ (b) $540 \div 1.5^2 = 240\ \text{cm}^2$

15 Simultaneous — line and curve

32 **SUBSTITUTE**

$x^2 + 2(x + 1)^2 = 22 \Rightarrow 3x^2 + 4x - 20 = 0$

33 **FACTORISE**

$(3x + 10)(x - 2) = 0$

ANSWER $x = 2,\ y = 3$ or $x = -\frac{10}{3},\ y = -\frac{7}{3}$

16 Functions

34 **(A)**

$g(-2) = 4,\ f(4) = 11$

35 **(B)**

$$y(x - 2) = x + 1 \Rightarrow yx - x = 2y + 1 \Rightarrow x = \frac{2y + 1}{y - 1}$$

36 **(C)**

$(2x + 3)^2 = 25 \Rightarrow 2x + 3 = \pm 5$

ANSWER (a) 11 (b) $h^{-1}(x) = \frac{2x + 1}{x - 1}$ (c) $x = 1$ or $x = -4$

17 Maximum volume

37 **(A)**
base $(24 - 2x)$ by $(24 - 2x)$, height x

38 **DIFFERENTIATE**
$V = 576x - 96x^2 + 4x^3$, $\dfrac{dV}{dx} = 576 - 192x + 12x^2$

39 **SET TO 0**
$12(x^2 - 16x + 48) = 0 \Rightarrow (x - 4)(x - 12) = 0$; $x = 12$ gives no box

ANSWER $x = 4$, $V = 4 \times 16^2 = 1024\ \text{cm}^3$

18 Vectors — straight-line proof

40 **(A)**
$\vec{OP} = \mathbf{a} + \frac{1}{3}(\mathbf{b} - \mathbf{a}) = \frac{1}{3}(2\mathbf{a} + \mathbf{b})$

ANSWER (b) $\vec{OQ} = 2\mathbf{a} + \mathbf{b} = 3\vec{OP}$: parallel and both pass through O, so O, P, Q are on a straight line.

19 Change the subject

41 **MULTIPLY OUT**
$pk - 5p = 3k + 2$

42 **k TERMS TOGETHER**
$pk - 3k = 5p + 2 \Rightarrow k(p - 3) = 5p + 2$

ANSWER $k = \dfrac{5p + 2}{p - 3}$

20 Trigonometric graph

43 **MIDDLE AND HEIGHT**
max 5, min 1: middle $c = 3$, height $a = 2$

44 **PERIOD**
one full wave in 180°, so $b = 360 \div 180 = 2$

ANSWER $a = 2$, $b = 2$, $c = 3$